

Seminar Nr. 2, Probability and Statistics

Classical Probability; Rules of Probability; Conditional Probability; Independent Events

Theory Review

Classical Probability: $P(A) = \frac{\text{nr. of favorable outcomes}}{\text{total nr. of possible outcomes}} = \frac{N_f}{N_t}$.

Mutually Exclusive Events: A, B m. e. (disjoint, incompatible) $\Leftrightarrow P(A \cap B) = 0$.

Rules of Probability:

$$P(\bar{A}) = 1 - P(A);$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B);$$

$$P(A \setminus B) = P(A) - P(A \cap B).$$

Conditional Probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$.

Independent Events: A, B ind. $\Leftrightarrow P(A \cap B) = P(A)P(B) \Leftrightarrow P(A|B) = P(A)$.

Total Probability Rule: $\{A_i\}_{i \in I}$ a partition of S , then $P(E) = \sum_{i \in I} P(A_i)P(E|A_i)$.

Multiplication Rule: $P\left(\bigcap_{i=1}^n A_i\right) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P\left(A_n \middle| \bigcap_{i=1}^{n-1} A_i\right)$.

1. The faces of a cube are painted each in a different color (the cube is transparent on the inside). Then the cube is broken into 1000 smaller, equally-sized cubes and one such cube is randomly picked. Find the probability of the following events:

- a) A: the cube picked has exactly three colored faces;
- b) B: the cube picked has exactly two colored faces;
- c) C: the cube picked has exactly one colored face;
- d) D: the cube picked has no colored faces.

2. (Pigeonhole Principle) A postman distributes n letters in N mailboxes. What is the probability of the event A: there are m letters in a given (fixed) mailbox ($0 \leq m \leq n$)?

3. (Breaking Passwords) An account uses 8-character passwords, consisting of letters (distinguishing between lower-case and capital letters) and digits. A spy program can check about 1 million passwords per second.

- a) On the average, how long will it take the spy program to guess your password?
- b) What is the probability that the spy program will break your password within a week?
- c) Same questions, if capital letters are not used.

4. (System Reliability) Compute the reliability of the system in Figure 1, if each of the five components is operable with probability 0.92, independently of each other.

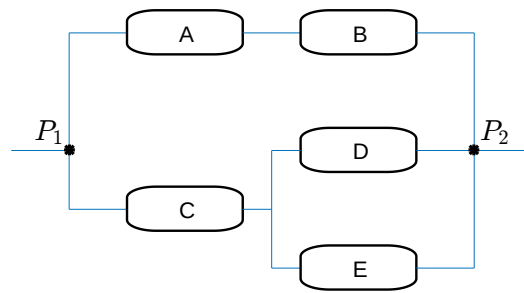


Figure 1: System in Problem 4

5. Among employees of a certain firm, 70% know Java, 60% know Python and 50% know both. What portion of programmers
 - a) does not know Python?
 - b) does not know Java and does not know Python?
 - c) knows Java, but not Python?
 - d) Are “knowing Java” and “knowing Python” independent of each other?
 - e) What is the probability that someone who knows Python, also knows Java?
 - f) What is the probability that someone who knows Java, does not also know Python?

6. Three shooters aim at a target. The probabilities that they hit the target are 0.4, 0.5 and 0.7, respectively. Find the probability that the target is hit exactly once (See Problem 4, in Seminar 1).

7. Under good weather conditions, 80% of flights arrive on time. During bad weather, only 50% of flights arrive on time. Tomorrow, the chance of good weather is 60%. What is the probability that your flight will arrive on time?

Bonus Problems:

8. There are 15 people waiting at a subway station. A train having 5 cars arrives and the passengers get in randomly. If each car has a capacity exceeding 15 persons, find the probability that at least one passenger will get into each car (event A).

9. There are two piles of used computer parts. All items in one pile are in working condition, while in the other pile, $1/4$ of the items are defective. An item is chosen randomly from one of the two piles and it is in working condition. If it is put back in the same pile, what is the probability that a second item, chosen at random *from the same pile* as the first one, will turn out to be defective?